CODE ACTIVITY

OUR TEAM

- Kundhana
- RaKshitha

DELIVERY ROUTE GRAPH CRUD – PACKAGE DELIVERY SERVICE

- Package delivery companies manage many locations and routes.
- Routes connect different cities or warehouses.
- Managing delivery routes manually is difficult.
- Data structures help organize and manage these routes efficiently.

PROBLEM STATEMENT

Delivery companies need a system to:

- Add new delivery routes
- Update route distances
- Delete unnecessary routes
- Search for routes quickly

This project provides a graph-based solution to manage delivery routes.

OBJECTIVE OF THE PROJECT

- Implement a real-world application using C programming
- Use Graph Data Structure
- Perform CRUD operations
- Create a menu-driven program
- Demonstrate delivery route management

DATA STRUCTURE USED

Graph Data Structure
Components:

- Vertices (Nodes) → Locations
- Edges → Routes
- Weight → Distance between locations

Example:
Chennai → Bangalore → Hyderabad

TECHNOLOGIES USED

- C Programming Language
- Graph Data Structure
- Structures (struct)
- Functions
- Menu-driven programming
- GitHub for project management

CRDU OPERATIONS

The program supports:

Create
- Add location
- Add route

Read
- Display routes

Update
- Modify route distance

Delete
- Remove route

PROGRAM FEATURES

- Add new delivery locations
- Create routes between locations
- Update route distance
- Delete routes
- Search routes
- Display all routes

Menu driven interface for easy use.

SAMPLE OUTPUT

1. Add Location
2. Add Route
3. Delete Route
4. Update Route
5. Search Route
6. Display Routes
7. Exit

Chennai → Bangalore : 350 Km

CONCLUSION

- Developed a Delivery Route Management System
- Used Graph Data Structure
- Implemented CRUD operations
- Demonstrated a real-world application using C programming

THANK YOU